

Practicum No.1**Date: 19/12/2025****Trait Assessment for CSE/AIML/DSCS/DS Founders****I. Practical Significance**

This practicum enables students to understand their own entrepreneurial personality traits and analyze how these traits align with potential engineering-based startup ventures. It introduces the concept of psychometric testing and entrepreneurial profiling, which help identify strengths, weaknesses, risk-taking abilities, creativity, and leadership potential essential for startup success. By undertaking a recognized entrepreneurial or psychometric test, students will learn how personality factors influence venture creation and how they can be matched with suitable CSE business opportunities such as AI/ML product development, cybersecurity solutions, IoT platform services, cloud-based application development, blockchain-based services, data analytics startups, mobile app solutions, and full-stack software product engineering. This exercise builds self-awareness, analytical thinking, and career readiness for innovation-driven entrepreneurship.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a pitch deck, project proposals for small/medium scale startups logically and scientifically.
- a) **Evaluate** founder readiness using computing-oriented entrepreneurial trait frameworks.(CDO).
- b) **Assess** innovation pathways in AI, data science, cloud computing, blockchain, and cybersecurity ventures.(CDO).
- c) **Interpret** competitive dynamics shaping CSE-focused high-tech and software-driven markets.(CDO).
- d) **Practice** professional ethics in data collection, system design, and user interaction within digital ecosystems.(ADO).

III. Relevant CO(s)

- **CO1. Identify** entrepreneurial traits and match them with suitable business opportunities.

IV. Practical Outcome (PrO)

- **PrO1** – Identify entrepreneurial and intrapreneurial traits based on **the given** test.

V. Related ADO(s) (keep as such)

- a) **Practice** professional ethics.
- b) **Function** effectively as a team member/leader.
- c) **Practice** integrity and honesty.

VI. Minimum Underpinning Theory

Entrepreneurship is not only a function of technical knowledge but also of personal traits and behavioural attributes that drive innovation, risk-taking, and opportunity recognition. A psychometric test helps assess these attributes scientifically. Common entrepreneurial traits include:

- **Innovativeness:** Ability to generate new ideas and apply technology creatively.
- **Risk-taking:** Willingness to undertake ventures under uncertainty.
- **Achievement Orientation:** Desire for excellence and measurable success.
- **Leadership:** Capacity to influence and guide others toward shared goals.
- **Resilience:** Ability to recover from setbacks and sustain motivation.
- **Decision-making:** Skill to evaluate options and select the most feasible one.

Mapping these traits to specific CSE ventures allows individuals to align their personality strengths with viable start-up opportunities. For instance:

- Highly innovative and problem-solving oriented students may excel in AI/ML product development startups or data science solution firms.
- Technically strong and detail-driven individuals may suit cybersecurity service ventures, cloud infrastructure solutions, or blockchain-based application startups.
- Creative and user-centric thinkers may thrive in UI/UX design studios, full-stack application development companies, or mobile app solution firms

Such mapping supports better career decisions, self-improvement, and entrepreneurial development.

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For accessing psychometric test websites	1 per student
2	Internet access	To complete online test and research	As required
3	Psychometric Test Platform	e.g., E-Scan, Entrepreneurial Mindset Test, etc.	1 per student
4	Worksheet / Excel	For recording and mapping results	1 per student

IX. Safety Precautions

- Ensure** data privacy while collecting survey responses.
- Avoid** falsifying or exaggerating personal traits.
- Follow** ethical standards in group discussions of personal results.

X. Procedure

- Participate** in psychometric or entrepreneurial aptitude test.
- Record** the obtained scores and personality profile.
- Identify** dominant entrepreneurial traits from the results.
- Prepare** a **Trait–Venture Mapping Table** relating each major trait to a specific startup domain.
- Discuss** results within the team and interpret findings.
- Present** a brief reflection report summarizing the overall entrepreneurial potential and scope for improvement.

XI. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	Personal laptop	Windows 11	1	Used for online test

XII. Actual Procedure Followed

- An online entrepreneurial psychometric test was completed individually.
- Test scores were recorded and dominant traits were identified.
- Traits were mapped with suitable CSE startup domains and discussed.

XIII. Observations and Data

S. No.	Entrepreneurial Trait Identified	Description / Strength Level	Suitable Startup Domain	Remarks
1	Innovativeness	High	AI/ML Product Development	Strong problem-solving
2	Risk Taking	Moderate	Startup MVP Development	Calculated risk
3	Leadership	High	Tech Startup Founder	Team coordination
4	Decision Making	High	Cloud & SaaS Solutions	Logical decisions
5	Resilience	Moderate	Cybersecurity Services	Handles pressure

XIV. Results/Observations

1. The student demonstrates strong innovativeness and leadership traits.
2. Entrepreneurial readiness is suitable for technology-driven startups.

XV. Interpretation of Results

The psychometric results indicate high creative thinking and leadership ability, making the student suitable for AI/ML, SaaS, and full-stack software ventures. Moderate risk-taking suggests a preference for structured startup environments.

Additionally, strong decision-making skills reflect the ability to analyze technical and business challenges logically, which is essential for product planning and technology selection in CSE-based startups. The presence of innovativeness shows readiness to adopt emerging technologies and convert ideas into practical digital solutions.

XVI. Conclusions

It is concluded that the student possesses key entrepreneurial traits such as innovativeness, leadership, and effective decision-making, which are essential for success in CSE-based startup ventures. The psychometric assessment highlights strong creative thinking and the ability to lead teams, making the student well-suited for technology-driven domains such as AI/ML solutions, SaaS platforms, and full-stack software development.

XVII. Suggested Practicum Related Questions

1. **Assess** the role of self-awareness in determining suitable CSE startup domains for individual entrepreneurial traits.
2. **Relate** specific entrepreneurial traits (e.g., risk-taking, innovativeness) to the potential success of AI/ML products, cybersecurity solutions, IoT platforms, cloud applications, blockchain services, data analytics startups, mobile app solutions, and full-stack software engineering.
3. **Evaluate** how psychometric test results can guide team composition and leadership assignment in technology start-ups.
4. **Identify** possible gaps in entrepreneurial skills revealed by psychometric assessments and propose strategies for skill enhancement.
5. **Examine** the influence of decision-making and resilience traits on the performance of early-stage CSE startups.
6. **Assess** the impact of ethical self-reporting on accurate trait mapping and long-term venture credibility.
7. **Justify** the selection of specific CSE startup domains based on an individual's dominant entrepreneurial traits.

8. **Examine** the role of communication and collaboration skills in the success of hardware/software integrated systems startups.
9. **Interpret** the significance of aligning personal innovation and achievement orientation with emerging technology trends in CSE.
10. **Evaluate** how awareness of entrepreneurial traits can improve opportunity recognition and risk management in IoT or semiconductor ventures.

XVIII References / Suggestions for further reading (keep as such)

1. Venture Meets Mission: Aligning People, Purpose, and Profit to Innovate and Transform Society, Arun Gupta, Gerard George, Thomas J. Fewer, Stanford University Press, Stanford, California, 2024, ISBN: 978-1503636283
2. So You Want to Start a Business: Eight Steps to Startup Success, John B. Vinturella, Business Expert Press, New York, USA, 2024, ISBN: 978-1637425855
3. Business Planning, Franklin A. Gevurtz, Foundation Press (West Academic), St. Paul, Minnesota, 2024, ISBN: 978-1685612764

XIX Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Completion of psychometric test	3	
2	Accuracy in recording and interpreting results	4	
3	Logical mapping of traits to ventures	4	
4	Team collaboration and discussion	3	
5	Ethical handling of personal data	4	
Product-related Skills: 12 Marks - 40%			
1	Clarity and presentation of trait-venture mapping table	3	
2	Interpretation of psychometric outcomes	3	
3	Quality of conclusions and reflections	3	
4	Timely submission and report quality	3	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	NEAVIS RISHVA. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Signature with Date
Process-Related skills (18)	Product-Related skills (12)	Total (30)		

